

E2E Automation Project:

Homburg Appointment System

Project Overview & Test Plan

Author: Valeriia Chuiko

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E2E Automation Project: Homburg Appointment System

Project Overview & Test Plan

1. Project Overview & Scope

- **System Under Test (SUT):** termine-reservieren.de/termine/homburg/.
- **Objective:** Designing an automated framework using Playwright.
- **Safety Policy:** Direct submission of real appointment reservations to the live municipal database at Step 6 is strictly restricted. All server responses for Step 4/5 time slots and final booking triggers are isolated via Playwright network interception (`page.route()`) to prevent production database pollution.

2. Testing Approaches

- **Manual Testing:** Exploratory, Functional, UI/UX Review, Accessibility Audit (BITV 2.0 / WCAG 2.1).
- **Automation Testing:** Functional E2E Testing via Playwright.

3. Test Environment & Stack

- **Browsers:** Chromium (Chrome/Edge), Firefox, WebKit (Safari).
- **Devices:** Desktop (1920x1080), Mobile (iPhone 14 Pro Max / 430px viewport).
- **Technical Stack:** Playwright (v1.40+), JavaScript (ES6+), Node.js (v20+), Git/GitHub.

4. Test Coverage (Key Modules)

- **Accessibility:** Audit of keyboard navigation (Tab, Enter, Space) and screen reader compatibility.
- **UI Components:** Interactive accordions and dynamic category lists.
- **Forms:** Input validation, data masking, and E-mail field protection.
- **Integrations:** Maps, geodata, and external navigation.
- **Session Logic:** Persistent countdown timers and predictive info blocks.

5. Out of Scope

- Backend database integrity testing & server-side penetration testing (Client-side input security, anti-paste restrictions, and deep-linking protection are fully covered).
- Performance & Load testing under high concurrent traffic.
- Physical navigation links leading outside the termine-reservieren.de domain.

Project Motivation & Rationale

Why this project? The choice of the termine-reservieren.de booking system is driven by the need to test a real-world, high-demand service that residents in Germany interact with daily. Working with this system provided not only a user-centric perspective but also the opportunity to perform an in-depth analysis of logical interface vulnerabilities.

Leveraging my background in manual testing, foundational ISTQB principles, and hands-on JavaScript knowledge, I designed a functional, helper-based automated Playwright framework tailored for a complex SPA (Single Page Application). The project focuses on covering the following critical mechanics:

- **Accessibility (a11y):** Manual audit (BITV 2.0 / WCAG 2.1), testing keyboard navigation, and ensuring screen reader compatibility.
- **UI Components:** Testing dynamic accordions, interactive maps (geotags, pop-ups), session elements (timers, predictive info blocks), slot selection filters, and multi-step registration forms.
- **Form Validation:** Validating complex inputs, handling data masks, and securing E-mail fields.

Key Skills Demonstrated Implementing this project allowed me to demonstrate proficiency in building robust, anti-flaky test logic, managing session states, utilizing API mocking, and applying deep knowledge of modern software accessibility standards.

Test Plan

Software Testing Lifecycle (STLC)

I. Requirements Analysis: Analyzing system logic, breaking down requirements into User Stories, and defining Acceptance Criteria using the BDD approach (Given-When-Then).

II. Manual Testing: Developing detailed test cases ready for automation and checklists for accessibility (BITV 2.0). Creating a Requirements Traceability Matrix (RTM) for 100% test coverage.

III. Test Automation Framework Engineering: Designing a scalable Playwright + JavaScript framework architecture. Developing stable E2E tests for positive, negative, and system scenarios, including API Mocking.

IV. CI/CD Integration, Execution & Reporting: Local and pipeline test execution. Gathering quality metrics and generating comprehensive reports. Conducting Root Cause Analysis (RCA) using logs, screenshots, and Playwright Trace Viewer.

V. Maintenance & Test Flakiness Management: Maintaining test documentation and updating the codebase in response to UI/UX changes. Optimizing locator stability and wait strategies to ensure high test reliability.

I. Requirements & Design

1.1 End-to-End User Flow & Step-by-Step Walkthrough: Detailed step-by-step description of the user's primary journey (Happy Path) from Step 1 through Step 6.

1.2 Technical & Interface Specifications: Technical rules covering session limits, field validation, UI states, and UX standards.

1.3 User Personas & Roles: Defined user profiles for conducting exploratory testing.

1.4 Functional Requirements (User Stories): User stories detailing all system functionalities.

1.5 Acceptance Criteria (Given-When-Then): Acceptance criteria defining the logic for each user story using the BDD format.

1.6 Test Scenarios (Positive & Negative): Segmentation of test cases into positive (Happy Path) and negative (Edge Cases) scenarios.

1.1 End-to-End User Flow & Step-by-Step Walkthrough

Step 1: Functional Unit Selection

- **Elements:** Bürgeramt (Citizens' Office), Fahrerlaubnis (Driver's License).
- **Action:** Selecting the relevant department.

Step 2: Service Selection

- **Logic:** Selecting services (maximum of 4 combined).
- **Navigation:** "Zurück zu den Funktionseinheiten" and "Weiter" buttons.
- **Overlay:** "Hinweis" window with "Ok", "Abrechnen", and "X" buttons.

Step 3: Branch Selection

- **Action:** Selecting a branch office via an interactive map.
- **Navigation:** "Zurück zu den Anliegen" and "Weiter" buttons.

Step 4: Time Slot Selection

- **Action:** Selecting a date from the calendar and a specific time slot; filtering functionality is available.
- **Navigation:** "Zurück zu den Standorten" button.
- **Overlay:** Confirmation window with "Ja", "Nein", and "X" buttons.

Step 5: Personal Data Entry

- **Required Fields (*) for Bürgeramt:** Vorname, Nachname, E-Mail, E-Mail-Wiederholung, Geburtsdatum, and "Ich willige ein" checkbox.
- **Required Fields (*) for Fahrerlaubnis:** Vorname, Nachname, E-Mail, E-Mail-Wiederholung, Geburtsdatum, Postleitzahl, Wohnort, and "Ich willige ein" checkbox.
- **Optional Fields:** Telefonnummer, Straße, Hausnummer.
- **Navigation:** "Zurück zu den Terminvorschlägen" and "Reservieren" buttons.

Step 6: Confirmation (Out of Scope)

- **Result:** Final screen confirming the reservation.
- **Isolation:** Email confirmation link delivery.

1.2 Technical & Interface Specifications

1.2.1 Constraints & Rules

- **Sequential Workflow:** Steps must be navigated strictly using "Back" / "Continue" buttons; direct URL access is restricted and triggers an error.
- **Global Service Limit:** Maximum of 4 services allowed per booking session.
- **Multi-Category Restriction:** Selecting services from different departments in one session is prohibited; changing the department at Step 1 clears the selected services cache.

- **Counter Logic:** The minus (-) button is disabled when the value is 0. The plus (+) button is disabled for all services in expanded accordions once the global limit of 4 services is reached.
- **Read-Only Summary:** The "Übersicht" panel is an expanded-by-default accordion displaying selected data in read-only mode; data editing within this panel is disabled.
- **Session Integrity:** Simultaneous operation in multiple browser tabs is prohibited; parallel booking attempts return a session error.
- **Single Slot Selection:** Selecting a new time slot automatically cancels any previously selected slot.
- **Reservation Hold:** Selected slots are held for the user for the duration of the 25-minute session timer.
- **Data Privacy Consent:** The "Reservieren" button remains disabled until the user activates the data processing consent checkbox.
- **Email Lifecycle Rule:** Booking confirmation is completed only after clicking the link provided via Email; this link is also used for cancellation or rescheduling.

1.2.2 Input & Validation

- **Security & Base Validation:**
 - **XSS/Injection:** HTML tags and script tags (<script>) are automatically blocked in all input fields.
 - **Trim:** Excess whitespace is automatically removed from the beginning and end of all text fields before data submission.
 - **Mandatory Indicators:** Fields required for submission are marked with an asterisk (*).
- **Field Specifications:**
 - **Name Fields (Vorname/Nachname):** Text fields (max 200 chars); letters (including umlauts), hyphens, apostrophes, and spaces allowed; special characters and digits prohibited.
 - **Email Validation:** Validated against the name@domain.tld mask. "E-Mail" and "E-Mail-Wiederholung" fields must match exactly. Paste functionality is blocked for the confirmation field.
 - **Phone Number (Telefon):** Optional field (max 20 chars); digits only.
 - **Date of Birth (Geburtsdatum):** Validated for numeric birth year format (4 digits). Non-numeric and special characters (e.g., "1995,") must be rejected by frontend validation logic.
 - **Address Fields (Straße/Hausnummer):** Optional fields allowing both letters and digits.
 - **Postleitzahl/Wohnort (Fahrerlaubnis):** Mandatory for driver's license applications. Postal code accepts exactly 5 digits; city requires a minimum of 3 characters.

1.2.3 UI States & Handling

1. Header Controls

- **Header Icons (Contrast, Einfache Sprache / Language Selection):**
 - **Default:** Elements are interactive (pointer cursor), transparent background, grey icon color.
 - **Hover:** Visual inversion of icon and background style upon hover.
 - **Dropdown (Click on 2nd icon):** Activates dropdown menu with "Deutsch" link.

- **Logo (City Logo):**
 - **Static:** Non-interactive image.
- 2. **Action Buttons & Navigation**
 - **Active:** Dark green fill, white text, pointer cursor.
 - **Hover:** Programmatic opacity reduction, resulting in a faded visual appearance.
 - **Disabled:** Triggered when step conditions are unmet; semi-transparent light green fill, grey text, not-allowed cursor.
- 3. **Accordions & Summary Panel**
 - **Service Categories:** Interactive by default (pointer cursor); toggles content visibility (Expanded / Collapsed).
 - **Filter & Time Selection:** Accordion logic implemented for filters; filtering synchronized with calendar view.
 - **Übersicht Panel (Summary):** Interactive header for toggling visibility; content is read-only with editing disabled.
- 4. **Counter Controls (Step 2)**
 - **Button (-):** Disabled when value is 0 (not-allowed cursor); Active when value ≥ 1 (pointer cursor).
 - **Button (+):** Active by default; Disabled when the global limit (4 services total) is reached.
- 5. **Visual Slot States (Step 4)**
 - **Available:** Interactive state (pointer cursor).
 - **Unavailable:** Deactivated state; booking blocked; hover and click events inactive.
- 6. **Validation Status Indicators**
 - **Error Marker [!]:** Triggered on blur if validation fails; static red triangle icon.
 - **Success Marker [✓]:** Triggered on blur upon successful validation; static green circle icon.
 - **Tooltips (?):** Grey circular icon; triggers overlay text on click.
- 7. **Modals & Overlays**
 - **Modal Trigger:** Blocking overlay prevents interaction with the background interface.
 - **Interactive Elements:** Primary action buttons ("Ok", "Schliessen") feature green fill with hover-opacity effects; secondary buttons and close [X] icon are interactive.
- 8. **Session Timer**
 - **Static Text:** Non-interactive element.
 - **Dynamic State:** String-based object updating every minute (countdown from 25 to 00).
- 9. **Inline & Footer Links**
 - **Default:** Green text with underline.
 - **Hover:** Opacity reduction.
 - **Overlay Trigger (Footer):** Activates modal window without page reload.
 - **Target Isolation:** Uses target="_blank" attribute for external document links.
- 10. **Checkboxes (Datenschutz Step 5)**
 - **Unchecked:** Default state; empty box.
 - **Checked:** User-activated trigger; validation passed.

1.2.4 Session & Navigation

- **Session Lifecycle:**
 - **Session Initiation:** The session is activated upon loading Step 1; a global 25-minute countdown timer is enabled in the page footer.
 - **Data Persistence:** Selected parameters (services, location) are maintained between steps 1-3 during navigation via "Back" / "Continue" buttons and page refreshes (F5).
 - **Temporary Slot Reservation:** Upon confirming a time slot at Step 4, the slot is temporarily locked to the session until the process is completed or the timer expires.
 - **Data Storage & Privacy:** Data entered at Step 5 is stored in session memory (client-side) and is automatically destroyed upon closing the browser tab.
- **Expiration Handling:**
 - **Visual Timer:** Persistent countdown timer displayed across all UI layers.
 - **Soft Timeout:** At 01:00 remaining, a blocking modal appears, allowing the user to extend the session to 25 by clicking "Close".
 - **Hard Timeout:** If no action is taken by 00:00, the session is annulled, the booking slot is released, and the user is redirected to Step 1.
- **Navigation Logic & Resilience:**
 - **Browser Back Button:** Pressing the browser "Back" button does not reset the session and returns the user to the previous step; data cached within the session is re-validated upon return from Step 5.
 - **Progress Tracking:** The progress bar accurately displays the current stage across all 6 steps.
- **System Integrity:**
 - **SSL/HTTPS Encryption:** All data transmission processes are protected by the TLS (HTTPS) encryption protocol.
 - **Offline Handling:** Intercepted API failures (e.g., HTTP 500 during calendar.js fetch) must gracefully block Step 4 rendering and trigger a system error modal "Fehler bei der Terminauswahl".

1.2.5 Inclusivity, UX & GDPR

- **Accessibility (BITV 2.0 / WCAG 2.1 Compliance):**
 - **Keyboard Navigation:** All interactive elements (buttons, slots, checkboxes) are accessible via the Tab key and activatable using Enter or Space.
 - **Accessibility Handling:** All interactive and graphical interface elements utilize technical aria-label attributes to ensure correct interpretation by screen readers.
 - **State Reporting:** Dynamic components (accordions, validation markers) communicate current system status via specialized ARIA properties.
 - **Focus Indication:** A clear, visible focus ring surrounds the active element and remains present when modal windows are opened.
 - **Contrast Toggle:** A high-contrast mode (Black/White) is available and persists throughout all step transitions.
 - **Color Contrast:** All text and icons maintain a minimum contrast ratio of 4.5:1.
- **Responsive Behavior:**
 - **Touch Targets:** The minimum touch area for mobile elements is 44x44 pixels.

- **Adaptive Layout:** For mobile viewports (430px / under 480px), the calendar and service lists reconfigure into a single vertical column; horizontal scrolling is strictly prohibited.
- **Mobile Summary View:** The service summary view is collapsible on smaller resolutions to ensure it does not obstruct primary fields.
- **User Experience & Feedback:**
 - **Visual Hierarchy:** Primary action buttons ("Weiter", "Reservieren") are visually dominant, utilizing high-contrast brand colors.
 - **Language Consistency:** The platform includes an "Einfache Sprache an" toggle at the top of the interface, providing an accessible version of the content for users with varying language proficiency or cognitive needs.
- **Privacy & GDPR (DSGVO):**
 - **Data Minimization:** The system requests only the minimum dataset required for identification.
 - **Privacy Links:** Direct links to "Impressum" and "Datenschutz" are available in the footer at every stage to ensure compliance with German law.

1.3 User Personas & Roles

1. **Bürger / Standard User:** A user booking an appointment independently via desktop or mobile device. The primary goal is to successfully navigate all steps, secure a time slot, and validate the email address. This user expects the session timer to function accurately and requires clear error guidance when inputting data.
2. **Accessible User:** A user with accessibility needs who navigates the interface solely via keyboard (Tab, Enter, Space) or uses screen readers. This persona verifies strict compliance with the BITV 2.0 standard.
3. **Sachbearbeiter / Clerk (Out of Scope):** An administrative employee processing confirmed bookings within the internal CRM/Backend system. This role is outside the scope of UI Black Box testing.

1.4 Functional Requirements (User Stories)

Role: Bürger / Standard User

- **US_01: Entry Point & Session Initialization (Pre-step)**
 - **Statement:** As a User, I want to navigate to the booking page via a direct URL so that I can begin the appointment process from the first step.
- **US_02: Department Selection (Step 1)**
 - **Statement:** As a User, I want to select a specific department (Bürgeramt or Fahrerlaubnis) so that I can immediately view the relevant services.
- **US_03: Multi-Service Selection (Step 2)**
 - **Statement:** As a User, I want to select multiple services simultaneously (up to 4) so that I can manage all tasks in a single visit.
- **US_04: Required Documents Information (Step 2 - Hinweis)**
 - **Statement:** As a User, I want to view a list of required documents so that I can fully prepare for my appointment.
- **US_05: Location Selection (Step 3)**
 - **Statement:** As a User, I want to view available branches on an interactive map so that I can select a suitable location and proceed to the calendar.
- **US_06: Time Slot Selection (Step 4)**

- **Statement:** As a User, I want to view an interactive calendar and filtering tools so that I can quickly find and secure a convenient appointment time.
- **US_07: Data Entry & Form Submission (Step 5)**
 - **Statement:** As a User, I want to complete the form with my contact details and submit the request so that the institution can register my appointment.
- **US_08: Booking Confirmation (Step 6)**
 - **Statement:** As a User, I want to receive an on-screen booking confirmation and an activation link via email so that I can complete the registration process.
 - **Status:** Out of Scope.
 - **Rationale:** Testing this stage requires performing a real database transaction, which would pollute the production environment with invalid records.
- **US_09: Data Privacy (GDPR / Datenschutz)**
 - **Statement:** As a User, I want to see a mandatory consent checkbox for data processing at Step 5 and have access to the privacy policy so that my rights are protected in compliance with GDPR.

Role: Accessible User

- **US_10: Interface Accessibility (Accessibility - BITV 2.0)**
 - **Statement:** As a User utilizing keyboard navigation or screen reader software, I want all interactive elements and system states to be semantically marked up so that I can independently complete the booking without using a mouse.

System Resilience & Non-Functional Requirements

- **SYS_01:** Session Timeout Warning (01:00)
- **SYS_02:** Session Expiration & Hard Timeout (00:00)
- **SYS_03:** Network Failure Handling (Offline Simulation)
- **SYS_04:** Parallel Booking Conflict Handling
- **SYS_05:** Backend Failure / API 500 Error Handling

Infrastructure & Responsiveness Requirements

- **INF_01:** Mobile Layout Adaptability (Viewport 430px)
- **INF_02:** Touch Target Sizing (44x44px)
- **INF_03:** Modal Overlay Isolation & Keyboard Trapping

1.5 Acceptance Criteria (BDD: Given-When-Then)

US_01: Entry Point & Session Initialization

- **AC_01.1: Successful Page Load**
 - **Given:** The user has the direct URL to the booking system.
 - **When:** The user navigates to <https://termine-reservieren.de/termine/homburg/>.

- **Then:** Step 1 page loads successfully, displaying the correct department selection interface.
- **AC_01.2: Deep Linking Protection**
 - **Given:** The user does not have an active booking session (cleared cache or new browser tab).
 - **When:** The user attempts to access an internal step URL directly (e.g., Step 2 URL).
 - **Then:** The system blocks access and displays an error page informing the user that direct navigation is not permitted.

US_02: Department Selection (Step 1)

- **AC_02.1: Automatic Navigation to Service List**
 - **Given:** The user is on the "Schritt 1 von 6" (Step 1 of 6) page.
 - **When:** The user clicks on either the "Bürgeramt" or "Fahrerlaubnis" button.
 - **Then:** The system redirects the user to "Schritt 2 von 6" (Step 2 of 6).

US_03: Multi-Service Selection (Step 2)

- **AC_03.1: Accordion Behavior**
 - **Given:** The user is on the "Schritt 2 von 6" (Step 2 of 6) page, and the list of service categories is displayed.
 - **When:** The user clicks on any category row.
 - **Then:** The category expands to display the nested list of services, tooltip icons (?), +/- buttons, and selection counters. Clicking again collapses the category, hiding the content.
- **AC_03.2: Counter Plus & Navigation Activation**
 - **Given:** All service counters are set to 0, and the "Weiter" button is disabled.
 - **When:** The user clicks the (+) button for any service.
 - **Then:** The counter for that service increases to 1, the (-) button becomes active, and the "Weiter" button transitions to the enabled state.
- **AC_03.3: Counter Minus & Navigation State**
 - **Given:** A service is selected (counter equals 1).
 - **When:** The user clicks the (-) button.
 - **Then:** The counter decreases to 0, and the (-) button becomes disabled. If the total counter for all services on the page becomes 0, the "Weiter" button is automatically disabled.
- **AC_03.4: Service Limit Enforcement (Lock)**
 - **Given:** The user has selected 3 services.
 - **When:** The user clicks the (+) button on any service.
 - **Then:** All (+) buttons on the page transition to the disabled state. The (-) buttons for selected services and the "Weiter" button remain active.
- **AC_03.5: Service Limit Enforcement (Unlock)**
 - **Given:** The user has selected 4 services from any combination of categories.
 - **When:** The user clicks the (-) button on any selected service.
 - **Then:** All (+) buttons on the page transition to the active state. The (-) buttons for selected services and the "Weiter" button remain active.

US_04: Required Documents Information (Step 2 - Hinweis)

- **AC_04.1: Overlay Rendering**
 - **Given:** The user has selected between 1 and 4 services on Step 2.
 - **When:** The user clicks the "Weiter" button.
 - **Then:** A modal window opens displaying the list of required documents and control elements: [X] (close), "Ok" and "Abrechnen" .

US_05: Location Selection (Step 3)

- **AC_05.1: Branch Display & Map Interaction**
 - **Given:** The user is provided with a choice of branches on the map.
 - **When:** The user clicks the "Karte anzeigen" button on a branch card.
 - **Then:** An interactive map with branch markers/pins is rendered below the card, enabled for panning and zooming.
- **AC_05.2: Popup Activation**
 - **Given:** The user is provided with a choice of branches on the map.
 - **When:** The user clicks on a blue branch marker-pin on the map.
 - **Then:** An information window opens with "Standort auswählen" and "Hierhin navigieren" buttons.
- **AC_05.3: Branch Selection**
 - **Scenario 1 (Selecting Location):** Given the info window is active, when clicking "Standort auswählen", then the location is saved and the user is redirected to Step 4.
 - **Scenario 2 (Navigation):** Given the info window is active, when clicking "Hierhin navigieren", then Google Maps opens in a new tab (target="_blank"), leaving the session active.

US_06: Time Slot Selection (Step 4)

- **AC_06.1: 3-Week Horizon Availability**
 - **Given:** The user navigates to Step 4.
 - **When:** The page loads completely, and the system renders the list of available days for a 3-week period.
 - **Then:** The first available day is automatically expanded, displaying the grid of active time slots (time tiles).
- **AC_06.2: Filter Interaction (Accordion Filter Behavior)**
 - **Given:** The "Vorschläge filtern" (Filter suggestions) panel is in a collapsed state.
 - **When:** Clicking "Vorschläge filtern", it expands with "Zeitbereich" and "Wochentag" accordions.
 - **Scenario 1 (Filtering):** Selecting parameters and clicking "Filtern" updates the calendar grid.
 - **Scenario 2 (Reset):** Clicking "Zurücksetzen" resets filters to defaults and updates the grid.
- **AC_06.3: No Available Slots Error Scenario**
 - **Status:** Blocked / Deferred
 - **Reason:** Full automation execution is blocked due to the inability to predict or seed zero-slot availability in the live production test environment without test database access.
 - **Given:** The user applies filter criteria returning zero available slots.
 - **When:** The filter query returns an empty result.

- **Then:** The calendar grid is hidden, an error message "Kein freier Termin verfügbar" is displayed, the "Weiter" button is disabled, and only the "Zurück zu den Standorten" button remains active.
- **AC_06.4: Time Selection Overlay Trigger**
 - **Given:** The user is on the Step 4 page.
 - **When:** The user clicks on an available dark-green time slot.
 - **Then:** A "Hinweis" modal window opens, blocking the main interface.
 - **Then:** Clicking "Ja" secures the slot and proceeds to Step 5; clicking "Nein" or [X] closes the modal without saving.

US_07: Data Entry & Form Submission (Step 5)

- **AC_07.1: Global Validation UI Mechanics**
 - **Given:** Step 5 page is loaded, and all mandatory fields are marked with an error indicator [!].
 - **When:** The user fills in a field with valid data and moves the focus (blur event).
 - **Then:** The error marker [!] instantly changes to the success marker [✓]. If the field data is completely deleted, the error marker returns.
- **AC_07.2: Anti-Paste Restriction & Mismatch**
 - **Given:** The "E-Mail" field contains a valid address.
 - **When:** The user attempts to paste text into "E-Mail-Wiederholung".
 - **Then:** The interface blocks the paste action. The success marker [✓] in the "E-Mail-Wiederholung" field activates only when the text is entered manually and matches the text in the first field exactly.
- **AC_07.3: Date of Birth Validation Logic (Geburtsdatum)**
 - **Given:** Date of birth fields are empty.
 - **When:** Leaving Tag/Monat empty and entering a valid 4-digit year in "Jahr".
 - **Then:** The system activates the success marker [✓] for the year input.
- **AC_07.4: Submit Button Lock State**
 - **Given:** The form is displayed.
 - **When:** Any mandatory field or checkbox displays an error marker [!].
 - **Then:** The "Reservieren" button is in a disabled state.
- **AC_07.5: Submit Button Activation**
 - **Given:** All mandatory fields are validly filled and consent is checked.
 - **When:** All error markers transition to green checkmarks [✓].
 - **Then:** The "Reservieren" button transitions to the active state.
- **AC_07.6: Backward Navigation Cache Retention**
 - **Given:** The user has reached Step 5.
 - **When:** Navigating backward using the "Zurück" button to steps 4, 3, or 2.
 - **Then:** Previously selected services, location, and time slot parameters remain cached and populated in the UI.
- **AC_07.7: Dynamic Übersicht & Step Indicator Synchronization**
 - **Given:** The user navigates through any step of the booking process.
 - **When:** The user advances or goes back a step.
 - **Then:** The Übersicht panel and progress indicator update dynamically to accurately reflect current choices and active step status.

US_08: Booking Confirmation (Step 6) Status: Out of Scope (Testing this stage requires real database transactions in the live city portal).

US_09: Data Privacy Compliance (GDPR / Datenschutz)

- **AC_09.1: Interface Blocking**
 - **Given:** The user is on the Step 5 page.
 - **When:** The consent checkbox is unchecked.
 - **Then:** The "Reservieren" button remains strictly disabled.
- **AC_09.2: Access to Information**
 - **Given:** The booking widget is active.
 - **When:** Clicking the "Datenschutzerklärung" link in the footer.
 - **Then:** Privacy policy opens in an isolated modal overlay without resetting the booking session.

US_10: Interface Accessibility (Accessibility - BITV 2.0)

- **AC_10.1: Keyboard Focus**
 - **Given:** Navigating through any step using the keyboard.
 - **When:** Pressing the Tab key.
 - **Then:** Interactive elements are focused sequentially with a high-contrast focus ring.
- **AC_10.2: Element Activation**
 - **Given:** An interactive element is focused.
 - **When:** Pressing Enter or Space.
 - **Then:** The target action executes correctly.
- **AC_10.3: Focus Trap in Modal Windows**
 - **Given:** A modal window (e.g., Datenschutz) is active.
 - **When:** Navigating via Tab key.
 - **Then:** Keyboard focus is trapped within the modal and closes on Escape key.
- **AC_10.4: Screen Reader Semantics**
 - **Given:** Dynamic components (accordions, counters) change state.
 - **When:** Component state changes.
 - **Then:** Corresponding ARIA attributes (aria-expanded, aria-disabled) update dynamically in the DOM.

System Resilience & Non-Functional Requirements (SYS_01 – SYS_06)

- **SYS_01 (AC_SYS_01): Session Timeout Warning (01:00)**
 - **Given:** The user is on Step 5 and session time elapses.
 - **When:** The countdown reaches 01:00 minute remaining.
 - **Then:** A warning modal appears allowing session extension.
- **SYS_02 (AC_SYS_02): Session Expiration & Hard Timeout (00:00)**
 - **Given:** The session warning modal expires without user interaction.
 - **When:** The countdown reaches 00:00.
 - **Then:** The session is annulled, form data is cleared, and the user is redirected to Step 1 (?rs).
- **SYS_03 (AC_SYS_03): Network Failure Handling (Offline Simulation)**
 - **Given:** Form entry is complete at Step 5.
 - **When:** Network connection drops during submission.

- **Then:** DOM data is preserved and a network availability error is displayed.
- **SYS_04 (AC_SYS_04): Parallel Booking Conflict Handling**
 - **Given:** Concurrent requests are sent for a single appointment slot.
 - **When:** Two independent instances attempt booking simultaneously.
 - **Then:** The first request succeeds; the second thread intercepts Conflict 409 and displays a slot-occupied message.
- **SYS_05 (AC_SYS_05): Backend Failure / API 500 Error Handling**
 - **Given:** Transitioning from Step 3 to Step 4.
 - **When:** calendar.js returns HTTP 500 Internal Server Error.
 - **Then:** Navigation to Step 4 is blocked, displaying an explicit error message instead of an unhandled blank screen.
- **SYS_06 (AC_SYS_06): Calendar Slot Mocking & Isolation**
 - **Given:** The user proceeds through Step 4 (Date Selection).
 - **When:** The calendar requests available slots from the server.
 - **Then:** Network interception via page.route() replaces live calendar data with deterministic test slots to ensure test stability and prevent holding real municipal appointments in the production database.

Infrastructure & Responsiveness Requirements (INF_01 - INF_03)

- **INF_01 (AC_INF_01): Mobile Layout Adaptability**
 - **Given:** Mobile context (iPhone / 430px viewport width).
 - **When:** Navigating through all booking steps.
 - **Then:** Zero horizontal scrolling occurs, and form inputs stack vertically into a single column.
- **INF_02 (AC_INF_02): Touch Target Sizing**
 - **Given:** Mobile interaction context.
 - **When:** Interacting with buttons, counters, and time tiles.
 - **Then:** Interactive elements meet the minimum target size of 44x44 pixels to prevent misclicks.
- **INF_03 (AC_INF_03): Keyboard Accessibility & Focus Control**
 - **Given:** Keyboard navigation mode across any step.
 - **When:** Pressing Tab, Enter, Space, or Escape keys.
 - **Then:** Focus order moves sequentially with a visible focus ring, and legal modals enforce focus traps.

1.6 Test Scenarios

1.6.1 Positive Scenarios

- **TS_01_A - E2E Flow: Bürgeramt**
 - **Description:** Execution of a positive end-to-end scenario from Step 1 through Step 5. Selecting the "Bürgeramt" department, exactly 1 service, a location in Homburg, and the first available time slot. Step 5 is completed with valid personal data.
 - **Expected Result:** All steps up to the 5th are completed successfully; the "Reservieren" button transitions to the "enabled" state to proceed to Step 6.
- **TS_01_B - E2E Flow: Fahrerlaubnis**

- **Description:** Execution of a positive end-to-end scenario from Step 1 through Step 5. Selecting the "Fahrerlaubnis" department, exactly 1 service, a location in Homburg, and the first available time slot. Step 5 is completed with valid personal data.
- **Expected Result:** All steps up to the 5th are completed successfully; the "Reservieren" button transitions to the "enabled" state to proceed to Step 6.
- **TS_02 - UI: Service Counter Limits**
 - **Description:** Testing the business logic for service selection at Step 2. Verifying counter accuracy, "Weiter" button activation, and compliance with the system-wide limit (max 4 services). The test concludes at Step 2.
 - **Test Scenarios:**
 - **Initial State:** Verify that the "Weiter" button is in the "disabled" state when all counters are 0.
 - **Activation (0 → 1):** Increase the counter for any service to 1. Verification: "Weiter" button becomes "enabled", and the (-) button becomes active.
 - **Reset (1 → 0):** Decrease the service counter to 0. Verification: (-) button becomes "disabled", and the "Weiter" button reverts to "disabled".
 - **Reaching the Limit (3 → 4):** Increase the total count of services to 4. Verification: All (+) buttons on the page are automatically locked ("disabled"), while (-) buttons and "Weiter" remain active.
 - **Recovery from Limit (4 → 3):** Click the (-) button when 4 services are selected. Verification: All (+) buttons on the page are unlocked ("enabled"), and the "Weiter" button remains active.
 - **Expected Result:** All interface elements (counters, +/- buttons, "Weiter" button) synchronously toggle their states ("enabled"/"disabled") in strict accordance with the number of selected services, preventing exceeding limits or submitting an empty booking.
- **TS_03 - UI: Filter Interactivity**
 - **Description:** Proceeding to Step 4. Verifying accordion expansion, default date values, and the activation of time/day-of-the-week checkboxes. Testing filter application with dynamic availability processing. The test concludes at Step 4 or proceeds to Step 5 depending on appointment availability.
 - **Expected Result:**
 - The "Vorschläge filtern" accordion expands successfully; date range fields are pre-populated with the current date (From) and the date 21 days later (To), adjusted to the department's working days.
 - Time interval and day-of-the-week checkboxes are interactive and toggle to the "checked" state upon clicking.
 - **Outcome A (Slots Found):** The time grid is displayed; clicking the first available slot triggers the "Hinweis" modal window, and successful confirmation redirects the user to Step 5.
 - **Outcome B (No Slots Found):** A valid system message "Kein freier Termin verfügbar" is displayed; the test concludes at Step 4.
- **TS_04 - Navigation: Backward & Cache Stability**
 - **Description:** Navigation backward from Step 5 through Steps 4, 3, and 2 using the "Zurück" button to verify state preservation and dynamic Übersicht updates.

- **Expected Result:**
 - **Return to Step 4:** Navigating back from Step 5 to Step 4 retains: Funktionseinheit, Anliegen, Standort and Termin. The Übersicht panel displays all four fields correctly.
 - **Return to Step 3:** Navigating back from Step 4 to Step 3 retains: Funktionseinheit, Anliegen, and Standort. The Termin selection is cleared from the Übersicht panel.
 - **Return to Step 2:** Navigating back to Step 2 retains only: Funktionseinheit. All subsequent selections (Anliegen, Standort, Termin) are cleared from the Übersicht panel.
- **TS_14 - UI: Global Header & Overview Consistency**
 - **Description:** Verifying the stability and correct rendering of global interface elements (Header, Footer).
 - **Expected Result:**
 - **Header:** The logo and two icons are displayed on all steps (1-6); elements remain present and accessible throughout every step of the process.
 - **Footer:** All links (Hilfe, Impressum, Datenschutz, etc.) trigger modal windows upon clicking. The "Schließen" button within each window correctly closes the popup.
- **TS_15 - UI: Dynamic Übersicht & Step Indicator**
 - **Description:** Verifying the correctness of the dynamic "Übersicht" block, data accuracy within it, and the synchronization of the Step Indicator with the current booking stage.
 - **Expected Result:**
 - **Step Indicator:** The progress indicator (1-5) visually highlights the currently active step corresponding to the page content.
 - **Übersicht (Dynamics):**
 - **Step 1:** The "Übersicht" block is hidden.
 - **Steps 2-5:** The block is present with the accordion expanded, displaying accurate information regarding selected services, location, and time.

1.6.2 Negative & Edge Case Scenarios

- **TS_06 - Filter: No Available Slots Handling**
 - **Status:** Blocked by Test Data.
 - **Reason:** Full implementation (filter reset + continuation) is blocked due to the inability to predict slot availability after filtering in the current test environment.
 - **Description:** A scenario where filter parameters at Step 4 return an empty result (no slots available). Verification of message display and filter reset mechanism to ensure scenario continuation.
 - **Expected Result:**
 - **Negative Outcome (No Slots):** The time grid is hidden, and the system message "Kein freier Termin verfügbar" is displayed. The filter accordion collapses, and the "Zurücksetzen" button becomes active.
 - **Filter Recovery:** Clicking the "Zurücksetzen" button unchecks active filter criteria and restores the full calendar slot grid.

- Scenario Continuation: Selecting the first restored slot and confirming via the "Ja" modal successfully redirects the user to Step 5.
- **TS_07 - Security: Deep Linking Protection**
 - **Description:** Attempting a direct URL navigation to an internal booking step (Step 5) without an active session.
 - **Expected Result:** The system blocks access to the form and displays an error message.
- **TS_08 - Security: Form Validation**
 - **Description:** Verification of negative validation, input field security, and the state of the "Reservieren" button at Step 5 for both "Bürgeramt" and "Fahrerlaubnis" processes.
 - **Expected Result:**
 - Real-time Feedback: Upon entering invalid data (e.g., incorrect email format, invalid year, or mismatching emails), the system must immediately trigger a visual error response:
 - **Error Message:** Display of a specific red error text (e.g., "Die Mailadressen stimmen nicht überein").
 - **Visual Indicator:** The field/icon status changes from "valid" (green) to "invalid" (red).
 - **Button State Logic:** The "Reservieren" button remains strictly disabled while mandatory fields are empty or contain invalid data.
 - The button becomes enabled only after all validation checks for mandatory fields are passed and the consent checkbox is checked.
- **TS_16 - System: Paste Restriction on Email Confirmation**
 - **Description:** Verifying that the email confirmation field ("Email wiederholen") strictly requires manual input by blocking the paste (Ctrl+V/Right-click) functionality.
 - **Expected Result:** The application must prevent any text from being pasted into the email confirmation field. The field must remain empty or unchanged upon a paste attempt, ensuring the user is forced to type the input manually. This restriction acts as a primary preventative measure before secondary validation (email mismatch) is triggered.
- **TS_09 - Session: Warning & Extension**
 - **Description:** Fast-forwarding client-side JavaScript session timers to reach the 01:00 warning threshold at Step 5, verifying client-side warning modal activation, overlay blocking, and form data retention.
 - **Expected Result:**
 - When the client-side session timer approaches the critical threshold (1 minute remaining), an "Ihre Sitzung läuft aus in 1 Minute" warning triggers on Step 5 and the warning modal opens.
 - Clicking the "Schliessen" button closes the modal window without resetting form inputs.
- **TS_10 - Session: Timeout Expiration**
 - **Description / Precondition:** The user is on Step 5 with a partially filled form. No interface interactions occur until the session timer (25 minutes) fully expires.
 - **Expected Result:** An automatic redirect to Step 1 occurs with a session reset parameter (URL: . . . /homburg/?rs). The Step 5 interface is completely

locked, entered data is erased, and the footer timer restarts from the initial 25 minutes (displaying 24 on-screen).

- **TS_11 - System: Network Drop Simulation**
 - **Status:** Deferred.
 - **Reason:** Risk of data integrity violations in the live system. Forcing a connection drop during the booking submission process may result in "phantom" records or orphaned sessions in the database.
 - **Description:** The user is on Step 5 and has completed the form. At this moment, the browser context is switched to "Offline" mode. An attempt is made to submit the form or interact with elements requiring server communication.
 - **Expected Result:** The interface remains stable, and data in the DOM is preserved. Upon attempting to submit the form, the system displays a valid notification regarding the lost network connection.
- **TS_12 - System: Parallel Booking Conflict**
 - **Status:** Deferred.
 - **Reason:** Test environment limitations (dependency on a shared backend). Executing parallel booking requests requires an isolated database with rollback capabilities after each run, which is not feasible without backend code intervention.
 - **Description:** Testing concurrent access to a single time slot by simultaneously sending requests from Step 5 to Step 6 using two independent Playwright instances.
 - **Expected Result:** The first request completes successfully (booking confirmed), while the second thread intercepts an API error (Conflict 409), and the interface displays a validation message to the user indicating that the selected slot is already occupied.
- **TS_13 - System: API 500 Error Handling**
 - **Description:** Verifying UI resilience when calendar data fails to load.
 - **Expected Result:** Navigation to Step 4 must be blocked, displaying a graceful error notification ("Fehler bei der Terminauswahl").
- **TS_17 - Error Message on Email Mismatch**
 - **Description:** Validation of the email confirmation logic by entering non-matching email addresses in the "E-Mail" and "E-Mail wiederholen" fields. The scenario proceeds through steps 1-4, then attempts to trigger validation by entering distinct email strings and shifting focus.
 - **Expected Result:** The system identifies the inconsistency and displays the specific error message: "Die Mailadressen stimmen nicht überein." in the designated error container.
- **TS_18 - System: Field Validation on Incomplete Submission**
 - **Description:** Verification of the form submission state when required fields in Step 5 are left incomplete. The scenario completes steps 1-4, navigates to step 5, and populates all mandatory personal data fields except for one (e.g., leaving the "Nachname" field blank).
 - **Expected Result:** The "Reservieren" button remains in a "disabled" state, preventing the user from proceeding to Step 6, as the validation requirements for mandatory fields are not met.

II: Test Execution & Strategy

- 2.1 UI/UX & Responsive Static Checklists
- 2.2 Accessibility Auditing Checklist (BITV 2.0 / WCAG 2.1)
- 2.3 Manual Test Cases
- 2.4 Defect Reporting & Severity Criteria
- 2.5 Definition of Done (DoD)
- 2.6 Requirements Traceability Matrix (RTM)
- 2.7 System Resilience & Interruption Scenarios
- 2.8 Infrastructure & Accessibility Coverage

2.1 UI/UX & Responsive Static Checklists

Block 1: Cross-Device Responsiveness (Mobile Context / Viewport 430px — iPhone 13/14)

Focus: Ensuring layout integrity, touch-target sizing, and interaction behavior on mobile screens (430px width).

[✓] PASSED | Vertical Input Layout (Step 5)

- **Requirement:** Verify that text fields (Vorname, Nachname) and the date-of-birth triad (Tag / Monat / Jahr) automatically stack into a single vertical column, preventing element overlapping.
- **Evidence:** Verified visually on 430px viewport (mobile breakpoint). All text inputs and date fields auto-stack vertically in a single column without overlapping or clipping.

[✓] PASSED | On-Screen Keyboard Interaction (Step 5)

- **Requirement:** Ensure that the activation of input fields (Vorname, Nachname, E-Mail) does not cause the layout to shift irreversibly, nor does the keyboard overlay the active field. The "Reservieren" button must remain accessible after the keyboard is dismissed.
- **Evidence:** Input field remains fully visible when tapped, layout restores cleanly when the keyboard is closed, and the "Reservieren" button remains accessible after dismissing the keyboard.

[✓] PASSED (Minor Deviation) | Touch Target Sizing

- **Requirement:** Verify that navigation buttons (counters +/-, Step 4 time tiles) meet the minimum physical size of 44x44 pixels to prevent accidental misclicks.
- **Evidence:** Checked time slots and accordion headers via inspector; measured element heights range from 35.6px to 44px (accordion headers inherit legacy jQuery UI layout at ~36px height).

[✓] PASSED | Vertical Navigation Stack

- **Requirement:** Ensure "Weiter" and "Zurück" buttons in the footer stack vertically on mobile, with the "Zurück" button positioned above the primary "Weiter" button.

- **Evidence:** Footer navigation buttons align in a single vertical column, with the "Zurück" button positioned cleanly above the primary "Weiter" button.

[✓] PASSED | Horizontal Scroll Control

- **Requirement:** Confirm the complete absence of horizontal scrolling across all 5 steps. Content must scale and adapt 100% to the 430px viewport width.
- **Evidence:** Content layout scales perfectly to the 430px container width without horizontal scrolling or visual clipping across all 5 steps.

[✓] PASSED | Summary Panel (Übersicht) Behavior

- **Requirement:** Verify that the "Übersicht" block remains consistently expanded and readable throughout all steps on mobile devices. Ensure that the block's content is clearly visible and does not overlap or obstruct any active input fields, particularly on Step 5.
- **Evidence:** The "Übersicht" block remains expanded and fully readable across all steps without overlapping form fields or blocking active inputs on Step 5.

Block 2: External Link Integrity & Modal Logic

Focus: Validation of footer links, modal window behavior, and session persistence.

[✓] PASSED | Modal Trigger

- **Requirement:** Verify that clicking "Impressum" and "Datenschutz" in the footer triggers the respective modal overlay without reloading the page.
- **Evidence:** Clicking "Impressum" and "Datenschutz" triggers modal overlays directly on screen without reloading the page.

[✓] PASSED | Content & Closing

- **Requirement:** Verify that the modal opens with the correct German text content and includes a functional "Close" button.
- **Evidence:** Legal modals display valid German text and close cleanly via the primary "Schließen" button and top-right close icon.

[✓] PASSED | External Links Logic

- **Requirement:** Ensure that any links inside these modals (e.g., links to main Homburg city website) open in a new browser tab.
- **Evidence:** Links inside modal content open in a new tab, leaving the active booking session open.

[✓] PASSED | Session Integrity

- **Requirement:** Verify that opening and closing these modals does not interrupt the booking process, reset the session timer, or clear selected services in the main tab.
- **Evidence:** Opening/closing modals maintains form input state and keeps the session countdown active without resetting.

Block 3: Brand Identity & Visual Hierarchy

[✓] PASSED | Action Buttons

- **Requirement:** Verify that primary action buttons ("Weiter", "Reservieren") are displayed in dark-green, while secondary/grey buttons ("Abrechnen", "Zurück") are visually distinct.
- **Evidence:** Primary action buttons ("Weiter", "Reservieren") are styled in dark green, while secondary buttons are displayed in grey, ensuring clear visual distinction.

[✓] PASSED | City Logo Stability

- **Requirement:** Verify that the Homburg city logo is static. Clicking it should trigger no action, page reload, or session reset.
- **Evidence:** Clicking the Homburg city logo produces no action, does not reload the page, and keeps the active session intact.

[✓] PASSED | Header Menus

- **Requirement:** Verify that interactive header buttons (e.g., language selection) correctly open and close their dropdown menus without disrupting the booking session.
- **Evidence:** Header dropdown menus toggle open and close properly upon click without disrupting the booking session or resetting user progress.

[✓] PASSED | Filter Reset Functionality (Step 4)

- **Requirement:** Verify that clicking the "Zurücksetzen" button clears all active filter criteria (time spans, days of the week) and restores the calendar view to its default state.
- **Evidence:** Clicking "Zurücksetzen" unchecks all active time/day checkboxes, resets filter parameters to default values, and re-renders the full calendar slot grid without reloading the page.

2.2 Accessibility Auditing Checklist (BITV 2.0 / WCAG 2.1)

Block 1: Keyboard Accessibility & Focus Handling

Focus: Verifying the capability to complete the entire booking process without the use of a mouse or trackpad.

[✓] PASSED | Tab Order Sequence

- **Requirement:** Navigate through all 6 steps using the Tab key. Ensure focus movement is strictly sequential (top-to-bottom, left-to-right) across all interactive elements (buttons, accordions, inputs, time slots) without unexpected jumps or skipped elements.
- **Evidence:** Tabbing through the form navigates sequentially top-to-bottom, left-to-right across all interactive elements without skipping controls or jumping unexpectedly.

[✓] PASSED | Focus Ring Visibility

- **Requirement:** Confirm that every interactive element displays a clear, high-contrast focus ring when highlighted via keyboard navigation. The focus indicator must remain visible at all times across all steps.
- **Evidence:** High-contrast focus indicators appear on every interactive element upon receiving keyboard focus and remain visible during navigation.

[✓] PASSED | Element Activation (Enter / Space)

- **Requirement:** Verify that all primary actions (expanding accordions (Step 2), selecting a time slot (Step 4), and toggling the consent checkbox (Step 5)) execute correctly upon pressing Enter or Space.
- **Evidence:** Pressing Enter or Space successfully expands accordions, selects time slot buttons, and toggles form checkboxes.

[✓] PASSED | Modal Focus Trap

- **Requirement:** When the "Datenschutzerklärung" modal is triggered, ensure the keyboard focus is strictly contained within the popup. Tab navigation must cycle only through elements inside the modal (the "Hier" link, "Schliessen" button, and close icon) without leaking focus to the background UI.
- **Evidence:** Keyboard focus is trapped within the open modal overlay, cycling exclusively through modal controls without shifting to background page elements.

[✓] PASSED | Escape Key Functionality

- **Requirement:** Verify that the Privacy Policy modal closes instantly upon pressing the Escape key.
- **Evidence:** Pressing the Escape key immediately closes the active modal overlay and returns focus to the main interface.

Block 2: Screen Reader Semantics & ARIA States

Focus: Ensuring interface accessibility for visually impaired users utilizing screen readers (e.g., NVDA, VoiceOver).

[✓] PASSED | ARIA Labels for Graphics

- **Requirement:** Verify that all purely graphical elements (language toggles, modal close icons, validation markers) are equipped with localized descriptive `aria-label` attributes (de-DE).
- **Evidence:** Inspector analysis confirms non-text visual icons, language toggles, and modal close buttons carry descriptive German `aria-label` attributes.

[✓] PASSED | Dynamic State Updates

- **Requirement:** Verify that when an accordion is expanded, its state in the DOM tree updates to `aria-expanded="true"`, and when service limit buttons (+) are disabled, they receive the native disabled attribute. This ensures assistive technologies correctly announce these state changes to the user.

- **Evidence:** DOM tree inspection confirms interactive states dynamically update aria-expanded attributes upon component expansion, and service counter buttons (+) correctly apply the disabled state upon reaching the selection limit.

Block 3: Contrast & Interface Scaling

Focus: Verifying readability and usability for users with visual impairments.

[✓] PASSED | Text Contrast Ratio

- **Requirement:** Use browser developer tools to verify that the contrast ratio between text (including fine print in the footer) and the background color is at least 4.5:1 (WCAG 2.1 Level AA standard).
- **Evidence:** Inspector contrast checks confirm text elements and footer fine print meet or exceed the 4.5:1 ratio required by WCAG 2.1 Level AA.

[✓] PASSED | Page Zoom (200%)

- **Requirement:** Increase the browser zoom level to 200%. The interface must remain readable; text content must not overlap, and all interactive elements (including navigation buttons in the footer) must remain fully accessible and functional.
- **Evidence:** Layout scales smoothly at 200% zoom without text truncation or element overlapping, keeping all interactive controls fully accessible.

2.3 Manual Test Cases

- **TC_01: E2E Integration Test: Portal Entry Point**
 - **Priority:** High
 - **Preconditions:** Browser cache and cookies are cleared; no active booking session exists.
 - **Steps:**
 1. Navigate to the Homburg city portal: <https://www.homburg.de>.
 2. In the main navigation menu, click on "Rathaus".
 3. From the dropdown list, select "Stadtverwaltung", then navigate to "Online-Dienste".
 4. Locate the graphical tile labeled "Termin mit Bürgerbüro/Führerscheinstelle vereinbaren" and click it.
 - **Expected Result:** The widget's landing page (Step 1 of 6) loads successfully in a new browser tab. The address bar displays the target URL: <https://termine-reservieren.de/termine/homburg/>.
- **TC_02: Boundary Value Validation: Date of Birth Triad (Step 5)**
 - **Priority:** High
 - **Preconditions:** Active session; user is on Step 5. Mandatory text fields (Vorname, Nachname, E-Mail, E-Mail-Wiederholung) are populated with valid test data. Privacy consent checkbox is checked. Date of birth fields (Tag, Monat, Jahr) are initially empty.
 - **Steps:**

1. Clear the "Jahr" field, enter an out-of-range year (e.g., 1895), and unfocus the field.
 2. Verify the status marker next to the "Jahr" field and the state of the "Reservieren" button.
 3. Clear the "Jahr" field, enter a valid birth year (e.g., 1995), and unfocus.
 4. Verify the status marker next to the "Jahr" field and the state of the "Reservieren" button.
- **Expected Result:**
 1. **Step 1-2 (Invalid Year):** The system identifies the year as being outside the valid range. A red error marker appears next to the "Jahr" input, form validation fails, and the "Reservieren" button becomes disabled.
 2. **Step 3-4 (Valid Year):** A green success marker appears next to the "Jahr" field. The interface permits submission without Day/Month values; all mandatory validations are satisfied, and the "Reservieren" button becomes active.
 - **TC_03: UX Isolation & Legal Modal Logic**
 - **Priority:** Medium
 - **Preconditions:** User is on Step 1 of the booking widget. The global session timer in the footer is active and counting down.
 - **Steps:**
 1. Scroll to the bottom of the page and click the "Datenschutzerklärung" link in the footer.
 2. Visually confirm the appearance of the modal window, the correctness of the German text, and the background overlay.
 3. Locate the "Hier" hyperlink within the modal text and click it.
 4. Return to the initial widget tab and click the "Schliessen" button or the close [X] icon.
 - **Expected Result:**
 1. **Step 2:** The legal information modal opens as an overlay. The background is dimmed, and interactions with the Step 1 department selection buttons are completely disabled.
 2. **Step 3:** The full privacy policy document opens in a new browser tab. The booking session and the modal window on the original tab remain active and state-preserved; the footer timer continues to count down without interruption.
 3. **Step 4:** The modal window closes smoothly. The user is returned to Step 1 in its initial state, and no previously selected parameters are reset.
 - **TC_04: Input Validation for 'Jahr' with Special Characters Acceptance**
 - **Priority:** High
 - **Preconditions:** Active session; user is on Step 5. Mandatory personal data fields are filled with valid test data. Privacy consent checkbox is checked.
 - **Steps:**
 1. Navigate to Step 5.
 2. Locate the "Jahr" input field.
 3. Enter a valid year followed by a special character (e.g., "1995,").
 4. Click outside the input field to trigger the blur event.

5. Observe the validation status marker next to the "Jahr" field and the state of the "Reservieren" button.
- **Expected Result:**
 1. The system must sanitize the input and reject any non-numeric characters.
 2. A red error marker must appear next to the "Jahr" field.
 3. The specific error message "Bitte geben Sie ein zulässiges Geburtsdatum ein" must be displayed.
 4. The "Reservieren" button must remain disabled.
 - **TC_05: Map Interaction & External Google Maps Navigation**
 - **Priority:** Medium
 - **Preconditions:** Active booking session; user is on Step 3.
 - **Steps:**
 1. Locate a branch card and click the "Karte anzeigen" button.
 2. Confirm that the interactive Leaflet map and blue marker pin render correctly below the card.
 3. Click on the blue branch marker pin on the map.
 4. Verify that the popup info window opens with two action buttons: "Standort auswählen" and "Hierhin navigieren".
 5. Click the "Hierhin navigieren" button.
 - **Expected Result:**
 1. Steps 1–2: The interactive map renders with zooming and panning enabled.
 2. Step 3–4: The info window opens smoothly over the marker pin.
 3. Step 5: Google Maps opens in a new browser tab (target="_blank") with direct route directions. The original booking tab remains active, keeping the session timer and selected options intact without page reload.

2.4 Defect Reporting & Severity Criteria

Defects discovered during manual or automated testing are classified by their impact on business logic (Severity) according to the following criteria:

Severity	Technical Criteria	Real-world Example (Homburg.de)
Blocker	Total failure of the Happy Path. No workaround available. UI results in "White Screen of Death" or infinite loading.	HTTP 500 error on Step 1. The "Weiter" button on Step 2 is unresponsive, making booking impossible.

Critical	Violation of core business logic or security. Essential features are blocked, despite potential complex workarounds.	The system allows form submission on Step 5 without checking the GDPR consent box. The session timer terminates prematurely.
Major	Functionality is operational but with significant deviations from UI standards or logic. The core business path remains viable.	Step 4 calendar displays/allows selection of dates from the previous month due to layout bugs. Filter reset function is fully broken.
Minor	Superficial UI defects, typos, or slight deviations from design mockups that do not impact the booking flow.	Grammatical error in a German label (e.g., lowercase "vorname"). The success icon [✓] is misaligned by 2 pixels.

Defect Reporting Workflow

When a defect is discovered that is not covered by automated tests, a formal Bug Report must be created. To ensure efficiency in the resolution process, every report must include the following elements:

- **Unique ID:** A reference identifier for tracking.
- **Summary:** A concise description of the issue.
- **Steps to Reproduce:** A step-by-step walkthrough to trigger the defect.
- **Actual Result:** A description of what is currently occurring.
- **Expected Result:** A description of the desired system behavior.
- **Mandatory Artifacts:** Evidence such as UI screenshots, screen recordings, or browser console logs (Trace Viewer logs for automated failures).

2.5 Requirements Traceability Matrix (RTM)

Req uire	Acce ptan ce	Description	Test Case / Method	Covera ge	Status	Linked Defect
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ment ID	Crite rion					
US_01	AC_0 1.1	Successful page load of the landing page	TS_01_A/B / TC_01	Covered (Auto & Manual)	Passed	-
US_01	AC_0 1.2	Deep linking protection	TS_07	Covered (Auto)	Passed	-
US_02	AC_0 2.1	Automatic navigation to Service list	TS_01_A/B, TS_02 / TC_01	Covered (Auto & Manual)	Passed	-
US_03	AC_0 3.1	Accordion UI behavior (expand/collapse)	TS_02, TS_01_A/B, Checklist 2.2 Block 2	Covered (Auto & Manual)	Passed	-
US_03	AC_0 3.2	Counter Plus & Navigation Activation (0 \$to\$ 1)	TS_02	Covered (Auto)	Passed	-
US_03	AC_0 3.3	Counter Minus & Navigation State (1 \$to\$ 0)	TS_02	Covered (Auto)	Passed	-
US_03	AC_0 3.4	Service Limit Enforcement (Locking (+) buttons)	TS_02	Covered (Auto)	Passed	-

US_03	AC_03.5	Service Limit Enforcement (Unlocking (+) buttons)	TS_02	Covered (Auto)	Passed	-
US_04	AC_04.1	Required documents modal overlay rendering	TS_01_A / TS_01_B	Covered (Auto)	Passed	-
US_05	AC_05.1	Branch display & interactive map rendering	TS_01_A / TS_01_B TC_05	Covered (Auto & Manual)	Passed	-
US_05	AC_05.2	Popup activation on branch marker-pin click	TS_01_A / TS_01_B TC_05	Covered (Auto & Manual)	Passed	-
US_05	AC_05.3	Location selection and external Google Maps navigation	TS_01_A / TS_01_B (Scen. 1) / TC_05	Covered (Auto & Manual)	Passed	-
US_06	AC_06.1	3-Week horizon availability calendar display	TS_03	Covered (Auto)	Passed	-
US_06	AC_06.2	Filter interaction (Accordion filter apply & reset)	TS_03 (Auto) / Checklist 2.1 Block 3 (Manual)	Covered (Auto & Manual)	Passed	-
US_06	AC_06.3	No available slots error	TS_06	Deferred / Blocked	Deferred	-

		scenario handling		(Live Data)		
US_06	AC_0 6.4	Time selection "Hinweis" modal window trigger	TS_03	Covered (Auto)	Passed	-
US_07	AC_0 7.1	Global validation UI mechanics	TS_08	Covered (Auto)	Failed	BUG_02 BUG_04
US_07	AC_0 7.2	Cross-field email verification & anti-paste restriction	TS_16, TS_17, TS_08	Covered (Auto)	Passed	-
US_07	AC_0 7.3	Date of Birth validation logic	TS_08, TS_01_A/B, TC_02, TC_04	Covered (Auto)	Failed	BUG_01
US_07	AC_0 7.4	Submit button lock state when mandatory fields are empty	TS_08, TS_18, TC_02, TC_04	Covered (Auto & Manual)	Passed	-
US_07	AC_0 7.5	Submit button activation when all fields are valid	TS_01_A/B, TS_08, TC_02	Covered (Auto & Manual)	Passed	-
US_07	AC_0 7.6	Backward Navigation &	TS_02, TS_09,	Covered (Auto)	Passed	-

		cache stability across steps	Checklist 2.1 Block 2			
US_07	AC_0 7.7	Dynamic Übersicht block & Step Indicator sync	TS_15, TS_04, Checklist 2.1 Block 1	Covered (Auto & Manual)	Passed	-
US_08	N/A	Booking Confirmation (Step 6)	Out of Scope	Out of Scope	-	-
US_09	AC_0 9.1	GDPR compliance mandatory checkbox interface blocking	TS_08, TS_18, TC_02	Covered (Auto & Manual)	Passed	-
US_09	AC_0 9.2	Isolated privacy policy modal window access	TS_14, TC_03, Checklist 2.1 Block 2	Covered (Auto & Manual)	Passed	-
US_10	AC_1 0.1–1 0.4	Keyboard Navigation & ARIA States Audit	Checklist 2.2 / TC_03	Covered (Manual)	Passed	-

2.6 System Resilience & Interruption Scenarios

Req uire ment ID	Acceptanc e Criterion	Description	Test Case / Method	Covera ge	Status	Linked Defect

SYS_01	AC_SYS_01	Session timeout warning (1:00 min)	TS_09	Covered (Auto)	Passed	-
SYS_02	AC_SYS_02	Session expiry & redirect (0:00)	TS_10	Covered (Auto)	Passed	-
SYS_03	AC_SYS_03	Network failure (Offline mode)	TS_11	Deferred	Deferred	-
SYS_04	AC_SYS_04	Parallel booking (Race condition)	TS_12	Deferred	Deferred	-
SYS_05	AC_SYS_05	Backend failure / API 500 error	TS_13	Covered (Auto)	Failed	BUG_03
SYS_06	AC_SYS_06	Calendar Slot Mocking & Isolation	TS_01_A /B, TS_03, TS_13	Covered (Auto)	Passed	-

2.7 Infrastructure & Accessibility Coverage

Requirement ID	Acceptance Criterion	Description	Test Case / Method	Coverage	Status	Linked Defect
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INF_01	AC_INF_01	Mobile layout & touch targets	Checklist 2.1 (Block 1)	Covered (Manual)	Passed	-
INF_02	AC_INF_02	Touch target sizing	Checklist 2.1 (Block 1)	Covered (Manual)	Passed	-
INF_03	AC_INF_03	Keyboard navigation (Tab order)	TC_03, Checklist 2.2 (Block 1)	Covered (Manual)	Passed	-

2.8 Definition of Done (DoD)

2.8.1 Manual Testing Done

[✓] **Requirement Coverage:** All User Stories (US_01-US_10) have been broken down according to business requirements; logical nuances (e.g., the date of birth triad) are documented; and each User Story is equipped with BDD-formatted (Given-When-Then) acceptance criteria that eliminate ambiguity and describe verifiable system behavior.

[✓] **Test Design:** Manual checklists have been designed for visual UI/UX inspections and responsiveness verification across mobile devices.

[✓] **Accessibility:** A manual interface audit has been conducted based on the BITV 2.0 standard (specifically verifying keyboard navigation using Tab, Enter, and Space).

[✓] **Traceability:** The Requirements Traceability Matrix (RTM) is complete, confirming that no business requirement remains untested.

[✓] **Risk & Resilience Analysis:** Negative scenarios and critical business risks (session timeouts, race conditions, network drops, and deep-linking protection) have been integrated into the testing architecture.

2.8.2 Automation Testing Done

[✓] **Code Coverage:** Automated functional and system scenarios have been fully implemented using Playwright and JavaScript (Node.js).

[✓] **Calendar Stability:** Flexible logic for identifying available dates has been implemented, ensuring tests remain stable when the calendar month rolls over (Anti-Flaky Logic).

[✓] **Production Isolation:** E2E automation validates the user flow from Step 1 to Step 5. Step 6 is explicitly excluded from execution to prevent submitting dummy data into the live Homburg production database.

[✓] **Cross-Browser Compatibility:** Tests execute in parallel across Chromium, Firefox, and WebKit. Automation successfully validated the Happy Path.

[✓] **Execution Stability:** In the event of a test failure, the framework automatically generates artifacts (screenshots and Trace Viewer logs) to facilitate rapid debugging.

[✓] **Sync & Review:** The requirements document has been cleared of implementation code, reviewed by the QA Lead, and approved as the "Single Source of Truth" for both manual and automated testing.

III. Test Automation Framework Engineering & Scripting

Focus: Designing a scalable framework based on Playwright + JavaScript. Development of robust E2E tests for positive, negative, and system scenarios utilizing API mocking.

3.1 Technology Stack Rationale

The booking widget (SUT) requires deterministic execution, tight time control, and strict network isolation. The Playwright + JavaScript stack was selected for the following architectural advantages demonstrated in the test suite:

- **Architecture & Speed:** Playwright communicates directly with browser engines (Chromium, Firefox, WebKit) via internal debugging protocols, enabling fast DOM operations and reliable multi-browser execution without WebDriver proxy overhead.
- **Auto-Waiting & Dynamic Readiness:** Playwright natively verifies visibility, actionability, and DOM states before executing clicks or inputs, eliminating fragile manual sleeps and ensuring reliable step transitions.
- **Virtual Time Control:** In session timeout tests Playwright's virtual clock API (`page.clock.fastForward`) fast-forwards client-side timers by 23+ minutes instantaneously without real-time waiting, allowing ultra-fast validation of timeout prompts and session resets.
- **API Interception & Mocking:** Built-in `page.route()` is leveraged to intercept network traffic:
 - Step 4: Intercepts the reservation request to prevent creating actual bookings in the live municipal database.
 - Resilience Scenarios: Mocks HTTP 500 responses on `calendar.js` to verify graceful client-side error handling.
- **Native DOM & Clipboard Emulation:** Features like `page.evaluate()` allow low-level event injection (e.g., custom `ClipboardEvent` instances in TS_16) to verify security restrictions like anti-paste controls on email confirmation fields.
- **Isolated Browser Contexts:** Each test executes in an independent `BrowserContext`, ensuring clean session states, isolated cookies, and zero cross-test state leakage.

- **JavaScript Efficiency:** Using a native Node.js/JavaScript environment ensures seamless integration with modern CI/CD pipelines (GitHub Actions), fast JSON handling for API mocks, low barrier for maintenance, and high execution speed during parallel runs.

🔴 ci: add playwright workflow for github actions
Playwright Tests #15: Commit 666042e pushed by LeraChuiko

main

🕒 1 minute ago
🔄 In progress

...

3.2 Directory & Project Structure

The project follows a streamlined structure designed for clarity and rapid maintenance. All functional test scripts, data definitions, and configuration files are consolidated within the tests/ directory to simplify navigation and project growth.

```

├── .github/
│   └── workflows/
│       └── playwright.yml      # CI/CD configuration manifest for GitHub Actions
├── assets/
│   ├── E2E_playwright.mp4      # Video recording of the automated end-to-end flow
│   ├── Test_execution_report_pw.png  # Screenshot of local Playwright test execution
│   └── UI_screenshots.pdf      # Documented interface layout screenshots
├── docs/
│   ├── Bug_report.pdf          # Formatted bug report for found defects
│   ├── Checklist.pdf           # Manual UI/UX and responsiveness checklist
│   ├── Manual_test_cases.pdf   # Step-by-step manual test cases
│   ├── Project_overview_test_plan.pdf # Master test plan and project strategy
│   ├── Test_execution_report.pdf # Test execution results and verdicts
│   └── Test_scenarios.pdf       # List of automated and manual test scenarios
├── tests/
│   ├── e2e_booking.spec.js      # End-to-end reservation workflow tests
│   ├── helpers.js              # Shared UI actions and locator utilities
│   └── session_timeout.spec.js  # Session timeout and warning prompt tests

```

— system_errors.spec.js	# Assertions for API failure handling
— system_resilience.spec.js	# Network disruption and recovery tests
— testData.json	# Test input data dictionary
— ui_components.spec.js	# Component tests (counters, maps, accordions)
— validation_security.spec.js	# Form validation and paste restriction tests
— .gitattributes	# Git configuration for line ending normalization (LF)
— .gitignore	# Git ignore rules for generated artifacts and logs
— .prettierignore	# Ignore rules for Prettier code formatting
— .prettierrc	# Prettier formatting configuration rules
— LICENSE	# Project distribution license (MIT)
— package-lock.json	# Locked node dependency versions
— package.json	# Project dependencies, metadata, and execution scripts
— playwright.config.js	# Playwright execution and context configuration
— README.md	# Main repository overview and setup guide

3.3 Architecture Note: Helper-Based Design

The framework utilizes a Helper-Based Modular Architecture rather than a standard Page Object Model.

- **Rationale:** By modularizing common UI actions into discrete, reusable helper functions, the framework achieves high agility.
- **Logic Isolation:** Locators are centralized in the top-level helper file, while complex business logic is encapsulated in functional helpers.
- **Maintenance:** This design allows for rapid modification of booking steps. Since the helpers are decoupled from specific test files, a single update in the helper library propagates across all test suites, ensuring high maintainability without the overhead of class-based POM structures.

3.4 Network Interception & Data Isolation

Instead of relying on live, unpredictable server state during slot selection, the framework utilizes Playwright's native network interception via `page.route()`.

- **Deterministic UI Verification:** On Step 4 (Date Selection), `page.route()` intercepts API calls to supply deterministic appointment slots. This eliminates test flakiness caused by shifting slot availability in the municipal calendar.

- **Infrastructure Fault Simulation:** For resilience testing (e.g., TS_13 - API 500 Error), `page.route()` intercepts calendar requests and returns HTTP 500 responses to verify that the UI gracefully handles backend failures without rendering a blank screen.
- **Production Isolation & Safety:** Utilizing network interception during slot selection, combined with marking Step 6 (Final Confirmation) as Out of Scope, guarantees that automated execution remains completely non-destructive and never creates real appointment holds in the City of Homburg production database.

IV. CI/CD: Implementation Status and Constraints

- **Objective:** I configured a CI/CD pipeline using GitHub Actions (`.github/workflows/playwright.yml`) to automate test execution on every commit.
- **Challenge:** During implementation, it became clear that the target website (`termine-reservieren.de`) actively blocks requests from cloud environments (like GitHub Actions), identifying them as suspicious bot activity.
- **Outcome and Resolution:** While the GitHub Actions workflow pipeline is fully written and operational, remote cloud runs remain unstable due to the target site's security restrictions. Therefore, primary test suite validation is maintained locally, where network conditions allow for complete, unblocked execution without false-negative failures.

V. Maintenance & Test Stability

- **Easy Maintenance:** I keep the code clean by using helper functions to separate test logic from UI selectors. If the website design changes, I only update one file instead of fixing dozens of tests.
- **Built-in Stability:** I rely on Playwright's auto-waiting features instead of manual pauses, which makes the tests faster and more reliable.
- **Smart Locators:** I prioritize robust locator strategies, utilizing semantic locators (`getByRole`, `getByLabel`) where available, alongside stable IDs and resilient CSS selectors (`#WeiterButton`, `.btn-number`) to ensure long-term test stability.
- **Root Cause Analysis:** When a test fails, I use Playwright Trace Viewer to see exactly what happened, allowing me to fix the underlying issue quickly rather than just guessing.